



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS



FACULTY OF COMPUTING
& INFORMATION SCIENCES

Egypt University of Informatics

Faculty of Computing and Information Sciences

Computer Vision (C-AI435)

Fourth year (2025-2026)

Computer Vision Project

1. Project Structure

- Duration: 5 weeks
- Team Size: 2–4 students

2. Deliverables

- Code: Well-documented GitHub repository with trained models.
- Report: 10–12 pages (abstract, Introduction, Related work, detailing methodology, and results)

3. Evaluation Criteria

Coding 50%	Creativity & Innovation 10%
Functionality & Accuracy 15%	Documentation 25%

4. Deadline

Discussion will be in the last week of the term

5. Suggested Project Domains

Project Code	Project Title	Description
CV-1	Medical Image Diagnosis/Segmentation	Classify or segment medical scans (X-rays, MRIs) to detect diseases.
CV-2	Crop Disease Detection	Identify plant diseases from images.
CV-3	Hand Gesture Recognition	Hand Gesture Recognition is a technology that detects, interprets, and understands hand gestures using computer vision and AI, allowing people to interact with computers, smart devices, or applications using hand movements instead of physical touch or voice.
CV-4	Recycling Automation	Waste-sorting system using image classification.
CV-5	Action Recognition in Sports	Classify actions like "goal," "foul," or "dunk" in sports videos.
CV-6	Lane Detection for Self-Driving Cars	Lane detection is a critical component of self-driving cars and autonomous vehicles. With it the vehicle will know where to go and avoid the risk of running into other lanes or getting off the road. This can prevent the driver/car system from drifting off the driving lane.
CV-7	Face Recognition Attendance System	Automate attendance using face recognition.
CV-8	Real-Time Object Detection & Tracking	Detect and track objects (e.g., cars, pedestrians) in video streams.
CV-9	Pose Estimation	Pose estimation involves identifying the location of specific points in an image, usually referred to as keypoints.
CV-10	Food-calorie-estimations	This system focuses mainly on the calculation of calories and other nutrients present in food.

For any other topics, please contact me to obtain my permission